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COMPRESSION SOCK WITH INTEGRATED DONNING AND DOFFING AID SYSTEM

Abstract

A compression sock may include a body comprising an inner face and an outer face, the inner face configured to interface with a wearer's anatomy, the body further comprising: a forefoot region, a midfoot region, a hindfoot region, a tendon region, a shin region; and a calf region. The compression sock may further include a plurality of sequential pulls coupled to the outer face of the body, at least one pull located proximal each region. Each pull may be configured to generate leverage for the wearer donning and doffing the compression sock.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-in-part of the pending International Application No. PCT/US2024/046925 filed Sep. 16, 2024, which claims the benefit of U.S. Provisional Patent Application No. 63/538,680 filed on Sep. 15, 2023, the contents of which are all incorporated herein by reference in their entireties.

TECHNICAL FIELD

[0002] The present disclosure generally relates to compression socks. More particularly, it relates to compression socks having an integrated donning and doffing aid system.

BACKGROUND

[0003] Medical grade compression socks are a staple treatment for multiple different medical issues. Most commonly, they are used by patients struggling with venous reflux resulting in lower leg swelling, skin changes and ulcerations. Venous reflux affects at least 40% of the population over the age of 65. Compression socks are highly effective in improving symptoms and mobility associated with venous disorders such as edema, phlebitis, and thrombosis. These elastic garments compress the limb, applying significant pressure to reduce vein diameter and increase blood flow. The goal of compression socks is to compress the surface veins, arteries, and muscles to narrow channels, increasing blood circulation throughout the body and preventing blood pooling in the lower limbs. However, given that the socks use much stronger elastics than basic everyday socks, these garments can be exceedingly difficult to put on and grip, especially for the elderly population and the medical professionals who assist them. This is especially so when the compression sock must be pulled over sharp angles, irregular shapes, and rigid protruding bones and tendons. Typical aids include rubberized gloves to aid in gripping the garment, wire structures to pre-stretch the garment and low friction sleeves to pull the sock over the heel. Some of these aids are incompatible with close-toed socks.

SUMMARY

[0004] In some aspects, the techniques described herein relate to a compression sock configured to be worn by a wearer, including: a body including an inner face and an outer face, the inner face configured to interface with a wearer's anatomy, the body further including: a forefoot region; a midfoot region; a hindfoot region; a tendon region; a shin region; and a calf region; and a plurality of sequential pulls coupled to the outer face of the body, at least one pull located proximal each region, and wherein each pull is configured to generate leverage for the wearer donning and doffing the compression sock.

[0005] In some aspects, the techniques described herein relate to a compression sock configured to be worn by a wearer, including: a body including an inner face and an outer face, the inner face configured to interface with a wearer's anatomy, the body further including: a forefoot region; a midfoot region; a hindfoot region; a tendon region; a shin region; and a calf region; wherein the outer face of the body defines a plurality of sequential openings, at least one opening located proximal each region, and wherein each opening is configured to generate leverage for the wearer donning and doffing the compression sock.

[0006] In some aspects, the techniques described herein relate to an integrated compression sock aid that optimizes the friction between the sock and the wearer's skin at key catch points along the wearer's leg, such that the compression sock is easy to don while still maintaining compression integrity after donning.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] A more complete understanding of the subject matter may be derived by referring to the detailed description and claims when considered in conjunction with the following figures, wherein like reference numbers refer to similar elements throughout the figures.

[0008] FIG. **1** illustrates a perspective view of a compression sock in accordance with an embodiment of the present disclosure.

[0009] FIG. **2** is a cross-section drawing of the anatomy of a typical human foot, as well as a partial view of a human tibia and fibula.

[0010] FIG. **3** is a drawing of the anatomy of a typical human leg.

[0011] FIG. **4** is a cross-section drawing of the posterior calf and tendon anatomy of a typical human leg.

[0012] FIG. **5** is a perspective view of a compression sock in accordance with an embodiment of the present disclosure.

[0013] FIG. **6** is a perspective view of the compression sock of FIG. **5**, as worn on both legs.

[0014] FIG. **7** is a perspective view of the compression sock of FIGS. **5** and **6**, as worn on both legs.

[0015] FIG. **8** is a perspective view of the compression sock of FIGS. **5-7**, as worn on both legs.

[0016] FIG. **9** is a perspective view of the compression sock of FIGS. **5-8**, as worn on both legs.

[0017] FIG. **10** is a perspective view of a compression sock in accordance with an embodiment of the present disclosure.

[0018] FIG. **11** is a perspective view of the compression sock of FIG. **10**, as worn on both legs.

[0019] FIG. **12** is a perspective view of the compression sock of FIGS. **10** and **11**, as worn on both legs.

[0020] FIG. **13** is a perspective view of the compression sock of FIGS. **10-12**, as worn on both legs.

[0021] FIG. **14** is a perspective view of the compression sock of FIGS. **10-13**, as worn on both legs.

[0022] FIG. **15** is a perspective view of a compression sock in accordance with an embodiment of the present disclosure.

[0023] FIG. **16** is a perspective view of a compression sock in accordance with an embodiment of the present disclosure.

[0024] FIG. **17** is a perspective view of a compression sock in accordance with an embodiment of the present disclosure.

[0025] FIG. **18** is a perspective view of the compression sock of FIG. **17**, as worn on both legs, in accordance with various embodiments.

[0026] FIG. **19** is a perspective view of a compression sock in accordance with an embodiment of the present disclosure.

[0027] FIG. **20** is a perspective view of the compression sock of FIG. **19**, as worn on both legs, in accordance with various embodiments.

[0028] FIG. **21** is a perspective view of a compression sock in accordance with an embodiment of the present disclosure.

[0029] FIG. **22** is a perspective view of the compression sock of FIG. **21**, as worn on both legs, in accordance with various embodiments.

[0030] FIGS. **23** and **24** are illustrations of a compression sock incorporating a sequential set of pulls configured as reinforced openings in the compression sock material.

DETAILED DESCRIPTION

[0031] The following detailed description is merely illustrative in nature and is not intended to limit the embodiments of the subject matter of the application and uses of such embodiments. As used herein, the words “exemplary” and “example” mean “serving as an example, instance, or

illustration.” Any implementation or embodiment described herein as exemplary, or an example is not necessarily to be construed as preferred or advantageous over other implementations. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, or the following detailed description.

[0032] FIG. 1 illustrates a perspective view of a compression sock in accordance with an embodiment of the present disclosure. Here, the compression sock **100** is shown in a lightly stretched condition to highlight details of the compression sock **100**, as opposed to an unworn condition (e.g., folded in a drawer). It will be understood by those of ordinary skill in the art that compression socks stretch to the shape and contours of a wearer's foot and/or leg, and that none of the figures described herein are intending to illustrate the compression sock **100** as it would appear on the body of ever wearer.

[0033] The compression sock **100** of FIG. 1 includes a body **101**. The body **101** includes an inner face and an outer face **103**. Generally, the inner face may be seen when the sock is turned inside out, given that the inner face is configured to interface with a wearer's anatomy when worn. The body **101** includes a forefoot region **102**, a midfoot region **104**, a hindfoot region **106**, a tendon region **108**, a shin region **110**, and a calf region **112**. These regions correspond to the regions of a typical human limb, the anatomy of which is described in further detail in FIGS. 2-4.

[0034] FIG. 2 is a cross-section drawing of the anatomy of a typical human foot, as well as a partial view of a human tibia and fibula.

[0035] As shown, a foot **200** includes a plurality of phalanges **212** and metatarsals **210**.

Colloquially, the phalanges **212** are the toes or digits of the foot **200**. The metatarsals **210** connect the phalanges **212** to the rest of the foot **200**, and help form the arches of the foot **200**, which are essential in both weight bearing and walking. To do so, the metatarsals are necessarily of prismoid shape. Moreover, the fifth metatarsal (e.g., most proximal the smallest toe) includes a tuberosity portion that appears as a rounded protrusion. This protruding portion of the metatarsal set presents the first challenge to anyone attempting to don a compression sock over the foot **200**.

[0036] Connected to the metatarsals is the cuboid **211**, an irregular cube-like shaped bone which helps provide the foot **200** with stability. Like the fifth metatarsal, the cuboid **211** also includes small bony prominence (tuberosity) that presents a challenge to donning a compression sock. Similarly, next to the cuboid **211** is the medial navicular **209**, which connects the ankle to the foot, and is itself an irregular pyriform bone having a large protuberance (another challenge). The medial navicular **209** and cuboid **211** are part of the midfoot.

[0037] As shown in FIG. 2, a human's tibia **204** and fibula **202** form an ankle mortise **206**, which itself consists of a medial malleolus **208** and lateral malleolus **213**. This region is colloquially referred to as the ankle. The ankle may also be considered part of the hindfoot. The irregular shape of this region, particularly the protruding angles of the medial malleolus **208** and lateral malleolus **213** present additional challenges to donning a compression sock.

[0038] FIG. 3 is an outline of a typical human leg and FIG. 4 is a cross-section drawing of the posterior calf and tendon anatomy of a typical human leg.

[0039] As shown in FIGS. 3 and 4, the leg **300** (**400**) demonstrates the variety of contours the compression sock (FIG. 1. **100**) must accommodate. A forefoot **310**, comprised of irregularly shaped toes and joints, is tapered. A tuberosity **311** (**411**) on the fifth metatarsal protrudes from this tapered shape, angling back into a curve at a midfoot **312**. A lateral malleolus **308** (**406**) and medial malleolus **408** (collectively, the ankle) appear as rounded protrusions at a hindfoot **313** (**413**). Similarly, a heel **309** (**402**) rounds out the hindfoot **313** (**413**). From the heel **309** (**402**), a compression sock may be pulled up the leg **300** at a substantially 90-degree angle. A tibial region **304** (e.g., shin region) extends from the hindfoot **313** to a tibial tuberosity **302** just below the knee **314**.

[0040] While the tibial region **304** (e.g., anterior shin) of the leg **300** may appear to be substantially planar, a posterior calf **306** (**412**) is not. The posterior calf **306** (**412**) may be slightly curved or

significantly arched, depending on the thickness of a wearer's calf muscle.

[0041] Likewise, in FIG. 4, the leg **400** includes a tendon connecting the posterior calf muscles to the heel bone, a strong fibrous cord known as the Achilles tendon **404**. The Achilles tendon **404** is the thickest tendon in the human body and is responsible for plantar flexion of the foot at the ankle. The Achilles tendon **404** is also irregularly shaped, being thicker at the middle of the calf and gradually thinning as it approaches the heel **402**.

[0042] Anyone attempting to put on or take off the compression sock (FIG. 1 **100**) must contend with each of these unique anatomical structures simultaneously. The compression socks described herein may include integrated compression sock aids that optimize the friction between the sock and the wearer's skin at key catch points (unique anatomical structures) of the wearer's leg, such that the compression sock is easy to don while still maintaining compression integrity after donning. As described herein, the compression sock fabric in contact with the wearer's skin may be released with enough force at key anatomical structures, enabling the rest of the sock to slide on easily. For example, the fabric may be released on the sagittal plane (mid-line) of the heel and ankle. In such embodiments, the pulls, grips, and/or openings, described herein, may be more effective when located at anterior and posterior anatomical regions than at medial and lateral anatomical regions. The more such pulls, grips, and/or openings release the fabric at a centerline of the heel/ankle, the easier the compression sock is to don.

[0043] FIG. 5 is a perspective view of a compression sock in accordance with an embodiment of the present disclosure. Moreover, FIG. 6 is a perspective view of the compression sock of FIG. 5, as worn on both legs.

[0044] The wearer may be a patient afflicted with venous reflux resulting in lower leg swelling, skin changes and ulcerations. The wearer may be a user attempting to improve symptoms and mobility associated with venous disorders such as edema, phlebitis, and thrombosis. The wearer may be an athlete or user engaged in athletic activities, or a preventative care everyday user, or any user attempting to improve blood flow throughout the lower extremities. Accordingly, a compression sock **500** may have a strength ranging from 10-20 mm/Hg, 20-30 mm/Hg, 30-40 mm/Hg, and/or 40-50 mm/Hg.

[0045] The body **501** includes an inner face and an outer face **503**. Generally, the inner face may be seen when the sock is turned inside out, given that the inner face is configured to interface with a wearer's anatomy when worn. Accordingly, the inner face may be a smooth uninterrupted surface configured to minimize irritation of the skin.

[0046] The plurality of sequential pulls may be a system to aid wearers (and those assisting them) in donning and doffing the compression sock **500**. As shown in the present embodiment of FIGS. 5 and 6, there are first through ninth sequential pulls **514a-i**.

[0047] In various embodiments, the plurality of sequential pulls **514a-i** may be coupled to the outer face **503**. In various embodiments, the plurality of sequential pulls **514a-i** may be optionally coupled to any part of the outer face **503**. The plurality of sequential pulls **514a-i** may be stitched, glued, tied, stuck, strapped, bound, clipped, grafted onto, or otherwise attached to the outer face **503**. In various embodiments, the plurality of sequential pulls **514a-i** may be in either a substantially vertical or substantially horizontal orientation relative the compression sock **500**. In various aspects, the plurality of sequential pulls **514a-i** may be cut into the outer face **503** and reinforced with stitching reinforcement or additional material around the incisions to form the pulls **514a-i**. Stitching reinforcement or additional material may be, for example, two overlying pieces of neoprene or low friction fabric. For example, the two overlying pieces of neoprene or low friction fabric may be stitched on either side (a proximal side and a distal side) of each of the plurality of sequential pulls **514a-i**. In various embodiments, the proximal side may be a top side and the distal side may be a bottom side, with the neoprene or low friction fabric of the proximal side overlaying the neoprene or low friction fabric of the distal side.

[0048] In various embodiments, the plurality of sequential pulls **514a-i** may be integrated into the

outer face **503** by, for example, three-dimensional (3D) manufacturing techniques. In various embodiments, each of the plurality of sequential pulls **514a-i** may be large enough to accommodate one or two fingers, or an aid hook, but small enough to avoid fabric bunching under footwear or clothing. In various embodiments, each of the plurality of sequential pulls **514a-i** may range in width between 0.1 inches (in)-0.5 in, 0.5 in-0.75 in, 0.75 in-1 in, 1 in-2 in, or 2-3 in wide. In various embodiments, each of the plurality of sequential pulls **514a-i** may range in thickness between 0.01 inches (in)-0.05 in, 0.05 in-0.10 in, 0.10 in-0.15 in, 0.15 in-0.2 in, 0.2 in-0.25 in, 0.25 in-0.3 in, or 0.3 in-0.5 in thick. In various embodiments, each of the plurality of sequential pulls **514a-i** may be 0.25 in thick. In various embodiments, the stitching reinforcement or additional material may range in width between 0.1 inches (in)-0.5 in, 0.5 in-0.75 in, 0.75 in-1 in, 1 in-2 in, or 2-3 in wide. In various embodiments, the stitching reinforcement or additional material may be 0.25 in wide.

[0049] In various embodiments, the plurality of sequential pulls **514a-i** may be placed along the body **501** of the compression sock **500** where the pulls **514a-i** will create the most leverage (e.g., mechanical advantage) for the wearer, assistant, or an aid to pull the sock on and take the sock off (e.g., donning and doffing).

[0050] For instance, the body **501** includes a forefoot region **506**, a midfoot region **504**, a hindfoot region **508**, a tendon region **510**, a shin region **502**, and a calf region **512**. These regions correspond to regions of a typical human limb, the anatomy of which was described above in reference to FIGS. 2-4.

[0051] Each region is located proximal to at least one pull of the plurality of sequential pulls **514a-i**. For example, in various embodiments, the forefoot region **506** of the body **501** is proximal a wearer's metatarsals. The midfoot region **504** is proximal a wearer's medial navicular and lateral cuboid. The hindfoot region **508** is proximal a wearer's medial malleolus, lateral malleolus, and heel. The tendon region **510** is proximal a wearer's Achilles tendon. The shin region **502** is proximal a wearer's medial tibial tuberosity and lateral tibial tuberosity. The calf region **512** is proximal a wearer's posterior calf.

[0052] As shown in FIGS. 5 and 6, a first pull **514a** is proximal the forefoot region **506**. The first pull **514a** may therefore generate leverage in donning and doffing the compression sock **500** over the wearer's metatarsals. This first pull **514a** is also useful given that the forefoot region **506** is harder to grasp due to sock bunching that might occur at the wearer's phalanges and metatarsals. A second pull **514b** is proximal the midfoot region **504** and specifically proximal the wearer's medial navicular, generating leverage in donning and doffing the compression sock **500** over the wearer's medial navicular. A third pull **514c** is also proximal the midfoot region **504**, and specifically proximal the wearer's lateral cuboid, generating leverage in donning and doffing the compression sock **500** over the wearer's lateral cuboid.

[0053] Likewise, a fourth pull **514d** is proximal the hindfoot region **508** and wearer's medial malleolus, generating leverage in donning and doffing the compression sock **500** over the wearer's medial malleolus. A fifth pull **514e** is also located proximal the hindfoot region **508**, and specifically proximate the wearer's lateral malleolus. The fifth pull **514e** may generate leverage in donning and doffing the compression sock **500** over the wearer's lateral malleolus. Being on either side of the ankle, the fourth pull **514d** and fifth pull **514e** are also proximate the heel, and together or on their own, may generate leverage in donning and doffing the compression sock **500** over the wearer's heel. Using both the fourth pull **514d** and fifth pull **514e** to generate leverage over the heel may be advantageous given that the heel represents the sharpest angular turn as the sock is pulled up the leg.

[0054] Similarly, a sixth pull **514f** is located proximal the tendon region **510** and may generate leverage in donning and doffing the compression sock **500** over the wearer's Achilles tendon. A seventh pull **514g** is located proximal the shin region **502** and wearer's medial tibial tuberosity and may generate leverage in donning and doffing the compression sock **500** over the wearer's medial

tibial tuberosity. Similarly, an eighth pull **514h** is also located proximal the shin region **502**, and specifically, the wearer's lateral tibial tuberosity, and may generate leverage in donning and doffing the compression sock **500** over the wearer's lateral tibial tuberosity.

[0055] Moreover, as shown, a ninth pull **514i** is located proximal the calf region **512** and may generate leverage in donning and doffing the compression sock **500** over the wearer's posterior calf.

[0056] Accordingly, as the wearer, assistant, or aid is pulling the sock **500** over the wearer's leg, the pulls **514a-i** may be manipulated in sequential order, thus generating sufficient leverage to pull the compression sock **500** over parts of the anatomy that challenge those with limited mobility and/or grip strength.

[0057] In various embodiments, the compression sock **500** may be made of any suitable compressive fabric, including elastane, cotton, microfiber, and the like, and composites thereof. In various embodiments, the plurality of sequential pulls **514a-i** may be made of a fabric such as polyester, nylon, elastane, and the like, and composites thereof, or any fabric suitable for the pulls to stretch to lay flat when not in use.

[0058] FIGS. **7** through **9** are each additional perspective views of the compression sock **500** of FIGS. **5** and **6**, as worn on both legs.

[0059] FIG. **10** is a perspective view of a compression sock in accordance with an embodiment of the present disclosure. Moreover, FIG. **11** is a perspective view of the compression sock of FIG. **10**, as worn on both legs.

[0060] The wearer may be a patient afflicted with venous reflux resulting in lower leg swelling, skin changes and ulcerations. The wearer may be a user attempting to improve symptoms and mobility associated with venous disorders such as edema, phlebitis, and thrombosis. The wearer may be an athlete or user engaged in athletic activities, or a preventative care everyday user, or any user attempting to improve blood flow throughout the lower extremities. Accordingly, a compression sock **600** may have a strength ranging from 10-20 mm/Hg, 20-30 mm/Hg, 30-40 mm/Hg, and/or 40-50 mm/Hg.

[0061] The compression sock **600** may include a body **601** defining first through n.sup.th sequential openings (e.g., plurality of sequential openings). The compression sock **600** may include a body **601** defining any number of sequential openings. The body **601** includes an inner face and an outer face **603**. Generally, the inner face may be seen when the sock is turned inside out, given that the inner face is configured to interface with a wearer's anatomy when worn. Accordingly, the inner face may be a smooth uninterrupted surface configured to minimize irritation of the skin.

[0062] In embodiments, the number of sequential openings is limited, as shown in FIG. **10**, so that openings are only provided over or adjacent to specific regions of the leg when compression sock **600** is worn. For example, the openings may only be provided over or adjacent to forefoot region **606**, midfoot region **604**, hindfoot region **608**, tendon region **610**, shin region **602**, and calf region **612**. This effectively limits the number of openings that may be formed, thereby limiting their placement to specific regions of the leg. If, for example, a large number of openings were to be placed all over compression **600**, that could be problematic for the wearer.

[0063] Specifically, the locations of the openings for compression sock **600** are dictated by both anatomical necessity as well as physics to maximally reduce the force required to don the compression sock **600**. The Law of Laplace dictates that the garment will be tightest over the narrowest anatomical areas, as expressed by the equation $P=T/r$, where P =Pressure exerted on the surface, T =Tension in the garment, and r =Radius of the curvature of the area being compressed.

[0064] Therefore, placing the openings as illustrated in FIG. **10** to release these areas significantly reduces the forces required to pull the garment into the proper position. The number of openings should be so limited so as to maintain the compression function of compression sock **600** (too many, or arbitrarily-placed openings would greatly weaken the compression sock, effectively rendering it unable to perform its function) and physical integrity-care must be taken to interrupt the textile of compression sock **600** as little as possible. Any additional openings will directly

compromise the overall strength of compression sock **600**, increasing the likelihood that it will rip. [0065] The use of too many openings (e.g., openings formed in locations that don't play a role in assisting in the donning or removal of compression sock **600**) will negatively affect the compressive capabilities of compression sock **600**. Additionally, an overabundance of access openings may lead to blistering to the skin. As noted in Moffatt, C. J., O'Hare, L., & Mercier, G. (2020). Risks and Contraindications of Medical Compression Treatment. *Phlebology*, 35(4), 235-244, the skin moves differently in areas with different compression. This differential causes shear forces between the different areas and can lead to edema and significant blistering.

[0066] The plurality of sequential openings may be a system to aid wearers (and those assisting them) in donning and doffing the compression sock **600**. As shown in the present embodiment of FIGS. **10** and **11**, there are first through ninth sequential openings **614a-i**.

[0067] In various embodiments, the outer face **603** may define the plurality of sequential openings **614a-i**. In various embodiments, the outer face **603** may optionally define the plurality of sequential openings **614a-i** at any part of the outer face **603**. In various embodiments, the plurality of sequential openings **614a-i** may be in either a substantially vertical or substantially horizontal orientation relative the compression sock **600**. In various aspects, the plurality of sequential openings **614a-i** may be cut into the outer face **603**. In various aspects, the plurality of sequential openings **614a-i** may be reinforced with stitching around the incisions that form the openings **614a-i**. In various embodiments, each of the plurality of sequential openings **614a-i** may be large enough to accommodate one or two fingers, or an aid hook, but small enough to avoid fabric bunching under footwear or clothing. In various embodiments, each of the plurality of sequential openings **614a-i** may range in width between 0.1 inches (in)-0.5 in, 0.5 in-0.75 in, 0.75 in-1 in, 1 in-2 in, or 2-3 in wide. In various embodiments, each of the plurality of sequential openings **614a-i** may be 0.75 in wide.

[0068] In various embodiments, the plurality of sequential openings **614a-i** may be placed along the body **601** of the compression sock **600** where the openings **614a-i** will create the most leverage (e.g., mechanical advantage) for the wearer, assistant, or an aid to pull the sock on and take the sock off (e.g., donning and doffing).

[0069] For instance, the body **601** includes a forefoot region **606**, a midfoot region **604**, a hindfoot region **608**, a tendon region **610**, a shin region **602**, and a calf region **612**. These regions correspond to regions of a typical human limb, the anatomy of which was described above in reference to FIGS. **2-4**.

[0070] Each region is located proximal to at least one pull of the plurality of sequential openings **614a-i**. For example, in various embodiments, the forefoot region **606** of the body **601** is proximal a wearer's metatarsals. The midfoot region **604** is proximal a wearer's medial navicular and lateral cuboid. The hindfoot region **608** is proximal a wearer's medial malleolus, lateral malleolus, and heel. The tendon region **610** is proximal a wearer's Achilles tendon. The shin region **602** is proximal a wearer's medial tibial tuberosity and lateral tibial tuberosity. The calf region **612** is proximal a wearer's posterior calf.

[0071] As shown in FIGS. **10** and **11**, a first opening **614a** is proximal the forefoot region **606**. The first opening **614a** may therefore generate leverage in donning and doffing the compression sock **600** over the wearer's metatarsals. This first opening **614a** is also useful given that the forefoot region **606** is harder to grasp due to sock bunching that might occur at the wearer's phalanges and metatarsals. A second opening **614b** is proximal the midfoot region **604** and specifically proximal the wearer's medial navicular, generating leverage in donning and doffing the compression sock **600** over the wearer's medial navicular. A third opening **614c** is also proximal the midfoot region **604**, and specifically proximal the wearer's lateral cuboid, generating leverage in donning and doffing the compression sock **600** over the wearer's lateral cuboid.

[0072] Likewise, a fourth opening **614d** is proximal the hindfoot region **508** and wearer's medial malleolus, generating leverage in donning and doffing the compression sock **600** over the wearer's

medial malleolus. A fifth opening **614e** is also located proximal the hindfoot region **608**, and specifically proximate the wearer's lateral malleolus. The fifth opening **614e** may generate leverage in donning and doffing the compression sock **600** over the wearer's lateral malleolus. Being on either side of the ankle, the fourth opening **614d** and fifth opening **614e** are also proximate the heel, and together or on their own, may generate leverage in donning and doffing the compression sock **600** over the wearer's heel. Using both the fourth opening **614d** and fifth opening **614e** to generate leverage over the heel may be advantageous given that the heel represents the sharpest angular turn as the sock is pulled up the leg.

[0073] Similarly, a sixth opening **614f** is located proximal the tendon region **610** and may generate leverage in donning and doffing the compression sock **600** over the wearer's Achilles tendon. A seventh opening **614g** is located proximal the shin region **602** and wearer's medial tibial tuberosity and may generate leverage in donning and doffing the compression sock **600** over the wearer's medial tibial tuberosity. Similarly, an eighth opening **614h** is also located proximal the shin region **602**, and specifically, the wearer's lateral tibial tuberosity, and may generate leverage in donning and doffing the compression sock **600** over the wearer's lateral tibial tuberosity.

[0074] Moreover, as shown, a ninth opening **614i** is located proximal the calf region **612** and may generate leverage in donning and doffing the compression sock **600** over the wearer's posterior calf.

[0075] Accordingly, as the wearer, assistant, or aid is pulling the sock **600** over the wearer's leg, the openings **614a-i** may be manipulated in sequential order, thus generating sufficient leverage to pull the compression sock **600** over parts of the anatomy that challenge those with limited mobility and/or grip strength.

[0076] In various embodiments, the compression sock **600** may be made of any suitable compressive fabric, including elastane, cotton, microfiber, and the like, and composites thereof. In various embodiments, the stitching reinforcing the plurality of sequential openings **614a-i** may be made of a fabric such as polyester, nylon, elastane, and the like, and composites thereof.

[0077] Openings **614a-i** may, in some embodiments, be reinforced openings to improve the structural integrity of compression sock **600**. To illustrated, FIGS. **23** and **24** are illustrations of a compression sock incorporating a sequential set of pulls configured as reinforced openings in the compression sock material.

[0078] FIGS. **12** through **14** are each additional perspective views of the compression sock **600** of FIGS. **10** and **11**, as worn on both legs.

[0079] FIG. **15** is a perspective view of a compression sock in accordance with an embodiment of the present disclosure. The compression sock **700** may include a body **701** and first through n.sup.th sequential pulls (e.g., plurality of sequential pulls). The body **701** includes an inner face and an outer face **703**. The plurality of sequential pulls may be a system to aid wearers (and those assisting them) in donning and doffing the compression sock **700**. As shown in the present embodiment of FIG. **15**, there are first through sixth sequential pulls **714a-f**.

[0080] In various embodiments, the plurality of sequential pulls **714a-f** may be coupled to the outer face **703**. The plurality of sequential pulls **714a-f** may be stitched, glued, tied, stuck, strapped, bound, clipped, grafted onto, or otherwise attached to the outer face **703**. In various embodiments, the plurality of sequential pulls **714a-f** may be in either a substantially vertical or substantially horizontal orientation relative the compression sock **700**. In various aspects, the plurality of sequential pulls **714a-f** may be cut into the outer face **703** and reinforced with stitching around the incisions to form the pulls **714a-f**. In various embodiments, each of the plurality of sequential pulls **714a-f** may be large enough to accommodate one or two fingers, or an aid hook, but small enough to avoid fabric bunching under footwear or clothing. In various embodiments, each of the plurality of sequential pulls **714a-f** may range in width between 0.1 inches (in)-0.5 in, 0.5 in-0.75 in, 0.75 in-1 in, 1 in-2 in, or 2-3 in wide. In various embodiments, each of the plurality of sequential pulls **714a-f** may range in thickness between 0.01 inches (in)-0.05 in, 0.05 in-0.10 in, 0.10 in-0.15 in, 0.15 in-0.2 in, 0.2 in-0.25 in, 0.25 in-0.3 in, or 0.3 in-0.5 in thick. In various embodiments, each of

the plurality of sequential pulls **714a-f** may be 0.25 in thick.

[0081] In various embodiments, the plurality of sequential pulls **714a-f** may be placed along the body **701** of the compression sock **700** where the pulls **714a-f** will create the most leverage (e.g., mechanical advantage) for the wearer, assistant, or an aid to pull the sock on and take the sock off (e.g., donning and doffing).

[0082] For instance, as shown, the body **701** includes a midfoot region **704**, a hindfoot region **708**, and a calf region **712**. These regions correspond to regions of a typical human limb, the anatomy of which was described above in reference to FIGS. 2-4.

[0083] Each region is located proximal to at least one pull of the plurality of sequential pulls **714a-f**.

[0084] For example, as shown, a first pull **714a** is proximal the midfoot region **704** and specifically proximal the wearer's medial navicular, generating leverage in donning and doffing the compression sock **700** over the wearer's medial navicular. A second pull **714b**, third pull **714c**, and fourth pull **714d** are proximal the hindfoot region **708**, and specifically proximal the wearer's ankle mortise, generating leverage in donning and doffing the compression sock **700** over the wearer's ankle mortise.

[0085] Likewise, a fifth pull **714e** and sixth pull **714f** are located proximal the calf region **712** and may generate leverage in donning and doffing the compression sock **700** over the wearer's calf.

[0086] FIG. 16 is a perspective view of a compression sock in accordance with an embodiment of the present disclosure. The compression sock **800** may include a body **801** and first through n.sup.th sequential openings (e.g., plurality of sequential openings). The body **801** includes an inner face and an outer face **803**. The plurality of sequential openings may be a system to aid wearers (and those assisting them) in donning and doffing the compression sock **800**. As shown in the present embodiment of FIG. 16, there are first through sixth sequential pulls **814a-f**.

[0087] In embodiments, the number of sequential openings is limited, as shown in FIG. 16, so that openings are only provided over or adjacent to specific regions of the leg when compression sock **800** is worn. This effectively limits the number of openings that may be formed, thereby limiting their placement to specific regions of the leg. If, for example, a large number of openings were to be placed all over compression sock **800**, that could be problematic for the wearer.

[0088] Specifically, the locations of the openings for compression sock **800** are dictated by both anatomical necessity as well as physics to maximally reduce the force required to don the compression sock **800**. The Law of Laplace dictates that the garment will be tightest over the narrowest anatomical areas, as expressed by the equation $P=T/r$, where P =Pressure exerted on the surface, T =Tension in the garment, and r =Radius of the curvature of the area being compressed.

[0089] Therefore, placing the openings as illustrated in FIG. 10 to release these areas significantly reduces the forces required to pull the garment into the proper position. The number of openings should be so limited so as to maintain the compression function of compression sock **800** (too many, or arbitrarily-placed openings would greatly weaken the compression sock, effectively rendering it unable to perform its function) and physical integrity-care must be taken to interrupt the textile of compression sock **800** as little as possible. Any additional openings will directly compromise the overall strength of compression sock **800**, increasing the likelihood that it will rip.

[0090] The use of too many openings (e.g., openings formed in locations that don't play a role in assisting in the donning or removal of compression sock **800**) will negatively affect the compressive capabilities of compression sock **800**. Additionally, an overabundance of access openings may lead to blistering to the skin. As noted in Moffatt, C. J., O'Hare, L., & Mercier, G. (2020). Risks and Contraindications of Medical Compression Treatment. *Phlebology*, 35(4), 235-244, the skin moves differently in areas with different compression. This differential causes shear forces between the different areas and can lead to edema and significant blistering.

[0091] In various embodiments, the outer face **803** may define the plurality of sequential openings **814a-f**. In various aspects, the plurality of sequential openings **814a-f** may be cut into the outer face

803. In various aspects, the plurality of sequential openings **814a-f** may be reinforced with stitching around the incisions that form the openings **814a-f**. In various embodiments, the plurality of sequential openings **814a-f** may be in either a substantially vertical or substantially horizontal orientation relative the compression sock **800**. In various embodiments, each of the plurality of sequential openings **814a-f** may be large enough to accommodate one or two fingers, or an aid hook, but small enough to avoid fabric bunching under footwear or clothing. In various embodiments, each of the plurality of sequential openings **814a-f** may range in width between 0.1 inches (in)-0.5 in, 0.5 in-0.75 in, 0.75 in-1 in, 1 in-2 in, or 2-3 in wide.

[0092] In various embodiments, the plurality of sequential openings **814a-f** may be placed along the body **801** of the compression sock **800** where the openings **814a-f** will create the most leverage (e.g., mechanical advantage) for the wearer, assistant, or an aid to pull the sock on and take the sock off (e.g., donning and doffing).

[0093] For instance, as shown, the body **801** includes a midfoot region **804**, a hindfoot region **808**, and a calf region **812**. These regions correspond to regions of a typical human limb, the anatomy of which was described above in reference to FIGS. 2-4.

[0094] Each region is located proximal to at least one opening of the plurality of sequential openings **814a-f**.

[0095] For example, as shown, a first opening **814a** is proximal the midfoot region **804** and specifically proximal the wearer's medial navicular, generating leverage in donning and doffing the compression sock **800** over the wearer's medial navicular. A second opening **814b**, third opening **814c**, and fourth opening **814d** are proximal the hindfoot region **808**, and specifically proximal the wearer's ankle mortise, generating leverage in donning and doffing the compression sock **800** over the wearer's ankle mortise.

[0096] Likewise, a fifth opening **814e** and sixth opening **814f** are located proximal the calf region **812** and may generate leverage in donning and doffing the compression sock **800** over the wearer's calf.

[0097] FIG. 17 is a perspective view of a compression sock in accordance with an embodiment of the present disclosure. Moreover, FIG. 18 is a perspective view of the compression sock of FIG. 17, as worn on both legs. The wearer may be a patient afflicted with venous reflux resulting in lower leg swelling, skin changes and ulcerations. The wearer may be a user attempting to improve symptoms and mobility associated with venous disorders such as edema, phlebitis, and thrombosis. The wearer may be an athlete or user engaged in athletic activities, or a preventative care everyday user, or any user attempting to improve blood flow throughout the lower extremities. Accordingly, a compression sock **10** may have a strength ranging from 10-20 mm/Hg, 20-30 mm/Hg, 30-40 mm/Hg, and/or 40-50 mm/Hg.

[0098] The compression sock **10** may include a body **11** and first through n.sup.th sequential pulls (e.g., plurality of sequential pulls). The compression sock **10** may include any number of sequential pulls. The body **11** may include an inner face and an outer face **13**. Generally, the inner face may be seen when the sock is turned inside out, given that the inner face is configured to interface with a wearer's anatomy when worn. Accordingly, the inner face may be a smooth uninterrupted surface configured to minimize irritation of the skin.

[0099] The plurality of sequential pulls may be a system to aid wearers (and those assisting them) in donning and doffing the compression sock **10**. As shown in the present embodiment of FIGS. 17 and 18, there are first through sixth sequential pulls **24a-f**.

[0100] In various embodiments, the plurality of sequential pulls **24a-f** may be coupled to the outer face **13**. In various embodiments, the plurality of sequential pulls **24a-f** may be optionally coupled to any part of the outer face **13**. The plurality of sequential pulls **24a-f** may be stitched, glued, tied, stuck, strapped, bound, clipped, grafted onto, or otherwise attached to the outer face **13**. In various embodiments, the plurality of sequential pulls **24a-f** may be in either a substantially vertical or substantially horizontal orientation relative the compression sock **10**. In various aspects, the

plurality of sequential pulls **24a-f** may be a plurality of rungs stitched to the outer face **13** and reinforced with stitching reinforcement or additional material to strengthen the attachment of the rung-like pulls **24a-f** to the outer face **13**.

[0101] In various embodiments, each of the plurality of sequential pulls **24a-f** may be large enough to accommodate one to four fingers, or an aid hook, but small enough to avoid fabric bunching under footwear or clothing. In various embodiments, each of the plurality of sequential pulls **24a-f** may range in width between 0.1 inches (in)-0.5 in, 0.5 in-0.75 in, 0.75 in-1 in, 1 in-2 in, 2-3 in, or 3-5 in wide. In various embodiments, each of the plurality of sequential pulls **24a-f** may range in thickness between 0.01 inches (in)-0.05 in, 0.05 in-0.10 in, 0.10 in-0.15 in, 0.15 in-0.2 in, 0.2 in-0.25 in, 0.25 in-0.3 in, 0.3 in-0.5 in, or 0.5-1 in thick. In various embodiments, each of the plurality of sequential pulls **24a-f** may be 0.5 in thick.

[0102] In various embodiments, the plurality of sequential pulls **24a-f** may be placed along the body **11** of the compression sock **10** where the pulls **24a-f** will create the most leverage (e.g., mechanical advantage) for the wearer, assistant, or an aid to pull the sock on and take the sock off (e.g., donning and doffing).

[0103] For instance, the body **11** includes a forefoot region **16**, a midfoot region **14**, a hindfoot region **18**, a tendon region **20**, a shin region **12**, and a calf region **22**. These regions correspond to regions of a typical human limb, the anatomy of which was described above in reference to FIGS. 2-4.

[0104] Each region is located proximal to at least one pull of the plurality of sequential pulls **24a-f**. For example, in various embodiments, the forefoot region **16** of the body **11** is proximal a wearer's metatarsals. The midfoot region **14** is proximal a wearer's medial navicular and lateral cuboid. The hindfoot region **18** is proximal a wearer's medial malleolus, lateral malleolus, and heel. The tendon region **20** is proximal a wearer's Achilles tendon. The shin region **12** is proximal a wearer's medial tibial tuberosity and lateral tibial tuberosity. The calf region **22** is proximal a wearer's posterior calf.

[0105] As shown in FIGS. 17 and 18, a first pull **24a** is proximal the midfoot region **14** and specifically proximal the wearer's lateral cuboid, generating leverage in donning and doffing the compression sock **10** over the wearer's lateral cuboid. This first pull **24a** is also useful given that the midfoot region **14** is harder to grasp due to sock bunching that might occur at the forefoot region **16** proximal the wearer's phalanges and metatarsals. A second pull **24b** is proximal the midfoot region **14** and specifically proximal the wearer's medial navicular, generating leverage in donning and doffing the compression sock **10** over the wearer's medial navicular. Likewise, a third pull **24c** is located proximal the hindfoot region **18**, and specifically proximate the wearer's lateral malleolus. The third pull **24c** may generate leverage in donning and doffing the compression sock **10** over the wearer's lateral malleolus. Additionally, a fourth pull **24d** is proximal the hindfoot region **18** and wearer's medial malleolus, generating leverage in donning and doffing the compression sock **10** over the wearer's medial malleolus. Being on either side of the ankle, the third pull **24c** and fourth pull **24d** are also proximate the heel, and together or on their own, may generate leverage in donning and doffing the compression sock **10** over the wearer's heel. Using both the third pull **24c** and fourth pull **24d** to generate leverage over the heel may be advantageous given that the heel represents the sharpest angular turn as the sock is pulled up the leg.

[0106] Similarly, a fifth pull **24e** is located proximal the tendon region **20** and may generate leverage in donning and doffing the compression sock **10** over the wearer's Achilles tendon. A sixth pull **24f** is located proximal the shin region **12** and wearer's medial tibial tuberosity and may generate leverage in donning and doffing the compression sock **10** over the wearer's medial tibial tuberosity.

[0107] Accordingly, as the wearer, assistant, or aid is pulling the sock **10** over the wearer's leg, the pulls **24a-f** may be manipulated in sequential order, thus generating sufficient leverage to pull the compression sock **10** over parts of the anatomy that challenge those with limited mobility and/or grip strength. Moreover, the width of the pulls **24a-f** and where each pull is attached at the outer

face **13** (e.g., the part of the compression sock **10** that releases) may aid in improving ease of donning the compression sock **10**.

[0108] In various embodiments, the compression sock **10** may be made of any suitable compressive fabric, including elastane, cotton, microfiber, and the like, and composites thereof. In various embodiments, the plurality of sequential pulls **24a-f** may be made of the same compressive fabric as the compression sock **10**. In other embodiments, the plurality of sequential pulls **24a-f** may be made of a different fabric from the compression sock **10**. In various embodiments, the plurality of sequential pulls **24a-f** may be made of any fabric, such as polyester, nylon, elastane, and the like, and composites thereof, or any fabric wherein a total deformation of the plurality of sequential pulls **24a-f** is substantially similar to that of the compression sock **10** fabric. In various embodiments, the plurality of sequential pulls **24a-f** may be made of any fabric that increases the compression sock's ability to stretch.

[0109] FIG. **19** is a perspective view of a compression sock in accordance with an embodiment of the present disclosure. Moreover, FIG. **20** is a perspective view of the compression sock of FIG. **19**, as worn on both legs. The wearer may be a patient afflicted with venous reflux resulting in lower leg swelling, skin changes and ulcerations. The wearer may be a user attempting to improve symptoms and mobility associated with venous disorders such as edema, phlebitis, and thrombosis. The wearer may be an athlete or user engaged in athletic activities, or a preventative care everyday user, or any user attempting to improve blood flow throughout the lower extremities. Accordingly, a compression sock **30** may have a strength ranging from 10-20 mm/Hg, 20-30 mm/Hg, 30-40 mm/Hg, and/or 40-50 mm/Hg.

[0110] The compression sock **30** may include a body **31** defining first through *n*.sup.th sequential grips (e.g., plurality of sequential grips). The compression sock **30** may include a body **31** defining any number of sequential grips. The body **31** includes an inner face and an outer face **33**. Generally, the inner face may be seen when the sock is turned inside out, given that the inner face is configured to interface with a wearer's anatomy when worn. Accordingly, the inner face may be a smooth uninterrupted surface configured to minimize irritation of the skin.

[0111] The plurality of sequential grips may be a system to aid wearers (and those assisting them) in donning and doffing the compression sock **30**. As shown in the present embodiment of FIGS. **19** and **20**, there are first through sixth sequential grips **44a-f**.

[0112] In various embodiments, the outer face **33** may define the plurality of sequential grips **44a-f**. In various embodiments, the outer face **33** may optionally define the plurality of sequential grips **44a-f** at any part of the outer face **33**. In various embodiments, the plurality of sequential grips **44a-f** may be in either a substantially vertical or substantially horizontal orientation relative the compression sock **30**. In various embodiments, excess material of the outer face **33** may define the plurality of sequential grips **44a-f**. In various aspects, the plurality of sequential grips **44a-f** may be cut into the outer face **33**. In various aspects, the plurality of sequential grips **44a-f** may be reinforced with stitching around the incisions in the outer face **33**.

[0113] In various embodiments, each of the plurality of sequential grips **44a-f** may be large enough to accommodate one to four fingers, or an aid hook, but small enough to avoid fabric bunching under footwear or clothing. In various embodiments, each of the plurality of sequential grips **44a-f** may range in width between 0.1 inches (in)-0.5 in, 0.5 in-0.75 in, 0.75 in-1 in, 1 in-2 in, 2-3 in, or 3-5 in wide. In various embodiments, each of the plurality of sequential grips **44a-f** may range in thickness between 0.01 inches (in)-0.05 in, 0.05 in-0.10 in, 0.10 in-0.15 in, 0.15 in-0.2 in, 0.2 in-0.25 in, 0.25 in-0.3 in, 0.3 in-0.5 in, or 0.5-1 in thick. In various embodiments, each of the plurality of sequential grips **44a-f** may be 0.5 in thick.

[0114] In various embodiments, the plurality of sequential grips **44a-f** may be defined along the body **31** of the compression sock **30** where the grips **44a-f** will create the most leverage (e.g., mechanical advantage) for the wearer, assistant, or an aid to pull the sock on and take the sock off (e.g., donning and doffing).

[0115] For instance, the body **31** includes a forefoot region **36**, a midfoot region **34**, a hindfoot region **38**, a tendon region **40**, a shin region **32**, and a calf region **42**. These regions correspond to regions of a typical human limb, the anatomy of which was described above in reference to FIGS. 2-4.

[0116] Each region is located proximal to at least one grip of the plurality of sequential grips **44a-f**. For example, in various embodiments, the forefoot region **36** of the body **31** is proximal a wearer's metatarsals. The midfoot region **34** is proximal a wearer's medial navicular and lateral cuboid. The hindfoot region **38** is proximal a wearer's medial malleolus, lateral malleolus, and heel. The tendon region **40** is proximal a wearer's Achilles tendon. The shin region **32** is proximal a wearer's medial tibial tuberosity and lateral tibial tuberosity. The calf region **42** is proximal a wearer's posterior calf.

[0117] As shown in FIGS. 19 and 20, a first grip **44a** is proximal the midfoot region **34** and specifically proximal the wearer's lateral cuboid, generating leverage in donning and doffing the compression sock **30** over the wearer's lateral cuboid. This first grip **44a** is also useful given that the midfoot region **34** is harder to grasp due to sock bunching that might occur at the forefoot region **36** proximal the wearer's phalanges and metatarsals. A second grip **44b** is proximal the midfoot region **34** and specifically proximal the wearer's medial navicular, generating leverage in donning and doffing the compression sock **30** over the wearer's medial navicular. Likewise, a third grip **44c** is located proximal the hindfoot region **38**, and specifically proximate the wearer's lateral malleolus. The third grip **44c** may generate leverage in donning and doffing the compression sock **30** over the wearer's lateral malleolus. Additionally, a fourth grip **44d** is proximal the hindfoot region **38** and wearer's medial malleolus, generating leverage in donning and doffing the compression sock **30** over the wearer's medial malleolus. Being on either side of the ankle, the third grip **44c** and fourth grip **44d** are also proximate the heel, and together or on their own, may generate leverage in donning and doffing the compression sock **30** over the wearer's heel. Using both the third grip **44c** and fourth grip **44d** to generate leverage over the heel may be advantageous given that the heel represents the sharpest angular turn as the sock is pulled up the leg.

[0118] Similarly, a fifth grip **44e** is located proximal the tendon region **40** and may generate leverage in donning and doffing the compression sock **30** over the wearer's Achilles tendon. A sixth grip **44f** is located proximal the shin region **32** and wearer's medial tibial tuberosity and may generate leverage in donning and doffing the compression sock **30** over the wearer's medial tibial tuberosity.

[0119] Accordingly, as the wearer, assistant, or aid is pulling the sock **30** over the wearer's leg, the grips **44a-f** may be manipulated in sequential order, thus generating sufficient leverage to pull the compression sock **30** over parts of the anatomy that challenge those with limited mobility and/or grip strength. Moreover, the width of the grips **44a-f** and where each grip is defined by the outer face **33** (e.g., the part of the compression sock **30** that releases) may aid in improving ease of donning the compression sock **30**.

[0120] In various embodiments, the compression sock **30** may be made of any suitable compressive fabric, including elastane, cotton, microfiber, and the like, and composites thereof. In various embodiments, the plurality of sequential grips **44a-f** may be made of the same compressive fabric as the compression sock **30**. In other embodiments, the plurality of sequential grips **44a-f** may be made of a different fabric from the compression sock **30**. In various embodiments, the plurality of sequential grips **44a-f** may be made of any fabric, such as polyester, nylon, elastane, and the like, and composites thereof, or any fabric wherein a total deformation of the plurality of sequential grips **44a-f** is substantially similar to that of the compression sock **30** fabric. In various embodiments, the plurality of sequential grips **44a-f** may be made of any fabric that increases the compression sock's ability to stretch.

[0121] FIG. 21 is a perspective view of a compression sock in accordance with an embodiment of the present disclosure. Moreover, FIG. 22 is a perspective view of the compression sock of FIG. 21, as worn on both legs. The wearer may be a patient afflicted with venous reflux resulting in lower

leg swelling, skin changes and ulcerations. The wearer may be a user attempting to improve symptoms and mobility associated with venous disorders such as edema, phlebitis, and thrombosis. The wearer may be an athlete or user engaged in athletic activities, or a preventative care everyday user, or any user attempting to improve blood flow throughout the lower extremities. Accordingly, a compression sock **50** may have a strength ranging from 10-20 mm/Hg, 20-30 mm/Hg, 30-40 mm/Hg, and/or 40-50 mm/Hg.

[0122] The compression sock **50** may include a body **51** defining first through *n*.sup.th sequential grips (e.g., plurality of sequential grips). The compression sock **50** may include a body **51** defining any number of sequential grips. The body **51** includes an inner face and an outer face **53**. Generally, the inner face may be seen when the sock is turned inside out, given that the inner face is configured to interface with a wearer's anatomy when worn. Accordingly, the inner face may be a smooth uninterrupted surface configured to minimize irritation of the skin.

[0123] The plurality of sequential grips may be a system to aid wearers (and those assisting them) in donning and doffing the compression sock **50**. As shown in the present embodiment of FIGS. **21** and **22**, there are first through fourth sequential grips **64a-d**.

[0124] In various embodiments, the outer face **53** may define the plurality of sequential grips **64a-d**. In various embodiments, the outer face **53** may optionally define the plurality of sequential grips **64a-d** at any part of the outer face **53**. In various embodiments, excess material of the outer face **53** may define the plurality of sequential grips **64a-d**. In various aspects, the plurality of sequential grips **64a-d** may be cut into the outer face **53**. In various aspects, the plurality of sequential grips **64a-d** may be reinforced with stitching around the incisions in the outer face **53**. In various embodiments, the plurality of sequential grips **64a-d** may be in either a substantially vertical or substantially horizontal orientation relative the compression sock **50**.

[0125] In various embodiments, each of the plurality of sequential grips **64a-d** may be large enough to accommodate one to four fingers, or an aid hook, but small enough to avoid fabric bunching under footwear or clothing. In various embodiments, each of the plurality of sequential grips **64a-d** may range in width between 0.1 inches (in)-0.5 in, 0.5 in-0.75 in, 0.75 in-1 in, 1 in-2 in, 2-3 in, or 3-5 in wide. In various embodiments, each of the plurality of sequential grips **64a-d** may range in thickness between 0.01 inches (in)-0.05 in, 0.05 in-0.10 in, 0.10 in-0.15 in, 0.15 in-0.2 in, 0.2 in-0.25 in, 0.25 in-0.3 in, 0.3 in-0.5 in, or 0.5-1 in thick. In various embodiments, each of the plurality of sequential grips **64a-d** may be 0.5 in thick.

[0126] In various embodiments, the plurality of sequential grips **64a-d** may be defined along the body **51** of the compression sock **50** where the grips **64a-d** will create the most leverage (e.g., mechanical advantage) for the wearer, assistant, or an aid to pull the sock on and take the sock off (e.g., donning and doffing).

[0127] For instance, the body **51** includes a forefoot region **56**, midfoot region **54**, a hindfoot region **58**, a tendon region **60**, a shin region **52**, and a calf region **62**. These regions correspond to regions of a typical human limb, the anatomy of which was described above in reference to FIGS. **2-4**.

[0128] Each region is located proximal to at least one grip of the plurality of sequential grips **64a-d**. For example, in various embodiments, the forefoot region **56** of the body **51** is proximal a wearer's metatarsals. The midfoot region **54** is proximal a wearer's medial navicular and lateral cuboid. The hindfoot region **58** is proximal a wearer's medial malleolus, lateral malleolus, and heel. The tendon region **60** is proximal a wearer's Achilles tendon. The shin region **52** is proximal a wearer's medial tibial tuberosity and lateral tibial tuberosity. The calf region **62** is proximal a wearer's posterior calf.

[0129] As shown in FIGS. **21** and **22**, a first grip **64a** is proximal the midfoot region **54**, generating leverage in donning and doffing the compression sock **50** over the wearer's midfoot region. This first grip **64a** is also useful given that the midfoot region **54** is harder to grasp due to sock bunching that might occur at the forefoot region **56** proximal the wearer's phalanges and metatarsals. A

second grip **64b** is proximal the shin region **52**, generating leverage in donning and doffing the compression sock **50** up and down the wearer's leg. Likewise, a third grip **64c** is located proximal the hindfoot region **58** and the tendon region **60**, and may generate leverage in donning and doffing the compression sock **50** over the wearer's heel and up the Achilles tendon. This may be advantageous given that the heel represents the sharpest angular turn as the sock is pulled up the leg. Similarly, a fourth grip **64d** is located proximal the calf region **62** and may generate leverage in donning and doffing the compression sock **50** over the wearer's posterior calf.

[0130] Accordingly, as the wearer, assistant, or aid is pulling the sock **50** over the wearer's leg, the grips **64a-d** may be manipulated in sequential order, thus generating sufficient leverage to pull the compression sock **50** over parts of the anatomy that challenge those with limited mobility and/or grip strength. Moreover, the width of the grips **64a-d** and where each grip is defined by the outer face **53** (e.g., the part of the compression sock **50** that releases) may aid in improving ease of donning the compression sock **50**.

[0131] In various embodiments, the compression sock **50** may be made of any suitable compressive fabric, including elastane, cotton, microfiber, and the like, and composites thereof. In various embodiments, the plurality of sequential grips **64a-d** may be made of the same compressive fabric as the compression sock **50**. In other embodiments, the plurality of sequential grips **64a-d** may be made of a different fabric from the compression sock **50**. In various embodiments, the plurality of sequential grips **44a-f** may be made of any fabric, such as polyester, nylon, elastane, and the like, and composites thereof, or any fabric wherein a total deformation of the plurality of sequential grips **64a-d** is substantially similar to that of the compression sock **50** fabric. In various embodiments, the plurality of sequential grips **64a-d** may be made of any fabric that increases the compression sock's ability to stretch.

[0132] In some aspects, the techniques described herein relate to a compression sock configured to be worn by a wearer. The compression sock may include a body having an inner face and an outer face. The inner face may be a smooth surface interfacing with a wearer's anatomy. The body may further include a forefoot region, a midfoot region, a hindfoot region, a tendon region, a shin region, and a calf region. The compression sock may further include a plurality of sequential pulls coupled to the outer face of the body. At least one pull may be located proximal each region. Moreover, each pull may generate leverage for the wearer donning and doffing the compression sock. Accordingly, the plurality of sequential pulls may aid in donning and doffing the compression sock.

[0133] In some aspects, the techniques described herein relate to a compression sock configured to be worn by a wearer. The compression sock may include a body having an inner face and an outer face. The inner face may be a smooth surface interfacing with a wearer's anatomy. The body may further include a forefoot region, a midfoot region, a hindfoot region, a tendon region, a shin region, and a calf region. The outer face of the body of the compression sock may define a plurality of sequential openings. At least one opening may be located proximal each region. Moreover, each opening may generate leverage for the wearer donning and doffing the compression sock. Accordingly, the plurality of sequential openings may aid in donning and doffing the compression sock.

[0134] In some aspects, the techniques described herein relate to a compression sock configured to be worn by a wearer. The compression sock may comprise regions of graduated compression. In some aspects, of the regions of graduated compression, the ankle region may comprise greater compression force (e.g., pressure) than other regions. Such graduated compression may be advantageous for wearers seeking medical-grade compression.

[0135] In some aspects, the techniques described herein relate to a compression sock configured to be worn by a wearer. The compression sock may comprise regions of progressive compression. In some aspects, of the regions of progressive compression, the calf region may comprise greater compression force (e.g., pressure) than other regions. Such progressive compression may be

advantageous for wearers seeking to improve calf muscle circulation and pumping for various purposes, such as for example, sports activity.

[0136] The inventor has contemplated that these systems could be used on any compression sequencing scheme.

[0137] The above is presented to describe specific attributes of this invention. The inventor has contemplated that these systems could be used within any garment that has stretch, compression, stiffness to the textile or is otherwise difficult to don. The inventor has contemplated that these systems could be used on compression socks, and/or garments other than compression socks, e.g., pneumatic and non-pneumatic (lymphedema decompression systems). This description is not meant to limit the scope of this disclosure.

[0138] The invention will be especially pertinent to those with grip strength dysfunction such as those with a history of stroke, rheumatoid arthritis, or the like. These systems will also be especially useful for anyone who has limited range of motion. These systems will also help any patients, including patients who have lost strength and flexibility due to age. These systems will also help caregivers who have difficulty gaining grip and leverage. Anyone who benefits from these types of garments will benefit from this invention. Anyone who aids another person who wears these types of garments will benefit from this invention.

[0139] The preceding detailed description is merely illustrative in nature and is not intended to limit the embodiments of the subject matter or the application and uses of such embodiments.

[0140] As used herein, the word “exemplary” means “serving as an example, instance, or illustration.” Any implementation described herein as exemplary is not necessarily to be construed as preferred or advantageous over other implementations. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, or detailed description.

[0141] The connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in an embodiment of the subject matter. In addition, certain terminology may also be used herein for the purpose of reference only, and thus are not intended to be limiting, and the terms “first”, “second” and other such numerical terms referring to structures do not imply a sequence or order unless clearly indicated by the context.

[0142] The foregoing description refers to elements or nodes or features being “connected” or “coupled” together. As used herein, unless expressly stated otherwise, “connected” means that one element is directly joined to (or directly communicates with) another element, and not necessarily mechanically. Likewise, unless expressly stated otherwise, “coupled” means that one element is directly or indirectly joined to (or directly or indirectly communicates with, electrically or otherwise) another element, and not necessarily mechanically. Thus, although the schematic shown in the figures depict one exemplary arrangement of elements, additional intervening elements, devices, features, or components may be present in an embodiment of the depicted subject matter.

[0143] While at least one exemplary embodiment has been presented in the foregoing detailed description, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or embodiments described herein are not intended to limit the scope, applicability, or configuration of the claimed subject matter in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing the described embodiment or embodiments. It should be understood that various changes can be made in the function and arrangement of elements without departing from the scope defined by the claims, which includes known equivalents and foreseeable equivalents at the time of filing this patent application.

Claims

1. A compression sock configured to be worn by a wearer, comprising: a body comprising an inner face and an outer face, the inner face configured to interface with a wearer's anatomy, the body comprising an elastic material configured to apply a compressive force to a foot and calf of the wearer, the body including: a forefoot region; a midfoot region; a hindfoot region; a tendon region; a shin region; and a calf region; and a plurality of sequential pulls coupled to the outer face of the body, at least one pull located proximal each region, and wherein each pull is configured to generate leverage for the wearer donning and doffing the compression sock.
2. The compression sock of claim 1, wherein the forefoot region is configured to be proximal a wearer's metatarsals, wherein a first pull is proximal the forefoot region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's metatarsals.
3. The compression sock of claim 2, wherein the midfoot region is configured to be proximal a wearer's medial navicular, wherein a second pull is proximal the midfoot region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's medial navicular.
4. The compression sock of claim 3, wherein the midfoot region is configured to be proximal a wearer's lateral cuboid, wherein a third pull is proximal the midfoot region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's lateral cuboid.
5. The compression sock of claim 4, wherein the hindfoot region is configured to be proximal a wearer's medial malleolus, wherein a fourth pull is proximal the hindfoot region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's medial malleolus.
6. The compression sock of claim 5, wherein the hindfoot region is configured to be proximal a wearer's lateral malleolus, wherein a fifth pull is located proximal the hindfoot region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's lateral malleolus.
7. The compression sock of claim 6, wherein the tendon region is configured to be proximal a wearer's Achilles tendon, wherein a sixth pull is located proximal the tendon region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's Achilles tendon.
8. The compression sock of claim 7, wherein the shin region is configured to be proximal a wearer's medial tibial tuberosity, wherein a seventh pull is located proximal the shin region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's medial tibial tuberosity.
9. The compression sock of claim 8, wherein the shin region is configured to be proximal a wearer's lateral tibial tuberosity, wherein an eighth pull is located proximal the shin region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's lateral tibial tuberosity.
10. The compression sock of claim 9, wherein the calf region is configured to be proximal a wearer's posterior calf, wherein a ninth pull is located proximal the calf region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's posterior calf.
11. The compression sock of claim 1, wherein each of the plurality of sequential pulls is 0.75 inches (in) wide.
12. The compression sock of claim 1, wherein each pull is 0.25 inches (in) thick.
13. The compression sock of claim 1, wherein the body is made of an elastane.

- 14.** The compression sock of claim 1, wherein each pull is made of nylon.
- 15.** The compression sock of claim 1, wherein each pull is made of an elastane.
- 16.** A compression sock configured to be worn by a wearer, comprising: a body comprising an inner face and an outer face, the inner face configured to interface with a wearer's anatomy, the body further comprising: a forefoot region; a midfoot region; a hindfoot region; a tendon region; a shin region; and a calf region, wherein the outer face of the body defines a plurality of sequential openings, at least one opening located proximal each region, and wherein each opening is configured to generate leverage for the wearer donning and doffing the compression sock.
- 17.** The compression sock of claim 16, wherein the forefoot region is configured to be proximal a wearer's metatarsals, wherein a first opening is proximal the forefoot region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's metatarsals, wherein the midfoot region is configured to be proximal a wearer's medial navicular and lateral cuboid, wherein a second opening is proximal the midfoot region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's medial navicular, wherein a third opening is proximal the midfoot region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's lateral cuboid.
- 18.** The compression sock of claim 17, wherein the hindfoot region is configured to be proximal a wearer's medial malleolus and lateral malleolus, wherein a fourth opening is proximal the hindfoot region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's medial malleolus, wherein a fifth opening is located proximal the hindfoot region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's lateral malleolus.
- 19.** The compression sock of claim 18, wherein the tendon region is configured to be proximal a wearer's Achilles tendon, wherein a sixth opening is located proximal the tendon region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's Achilles tendon, wherein the shin region is configured to be proximal a wearer's medial tibial tuberosity, wherein a seventh opening is located proximal the shin region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's medial tibial tuberosity.
- 20.** The compression sock of claim 19, wherein the calf region is configured to be proximal a wearer's posterior calf, wherein a ninth opening is located proximal the calf region and is configured to generate leverage for the wearer donning and doffing the compression sock over the wearer's posterior calf.
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